

5.7 Digging Deeper Into Unit Rates

Lesson Introduction

 Benchmarks	MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate. MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.		 Learning Targets	- I can determine a unit rate when given different ratios. - I can interpret unit rates in real-world contexts. - I can use unit rates to solve problems.
 Benchmark Coherence	Building On		Working Toward	
	N/A		MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions. MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.	
 Essential Questions	- How do you determine a unit rate when given different ratios? - How do you interpret unit rates in real-world contexts? - How do you use unit rates to solve problems?			
 Lesson Narrative	Students continue exploring unit rates in a variety of contexts, including carnival tickets (5.7.1 Warm-Up), working and job pay (5.7.2 Guided Practice), and household purchases (5.7.4 Collaboration). Opportunity for differentiation in both content and process exist throughout the lesson, with students using tape diagrams explicitly during 5.7.2 Guided Instruction before utilizing strategies of three-column methods beyond. 5.7.4 Collaboration and 5.7.5 Bring It Together provide a variety of entry points for application and synthesis before formative assessment in 5.7.6 Wrap-Up.			
 Component Summary	5.7.1 Warm-Up (~10 min.)	Compare the unit rates of carnival tickets (<i>SE p. 204</i>)	5.7.4 Collaboration (10-15 min.)	Collaboratively interpret and use unit rates to solve real-world problems (<i>SE p. 207</i>)
	5.7.2 Guided Instruction (10-20 min.)	Use tape diagrams to interpret unit rates (<i>SE p. 204</i>)	5.7.5 Bring It Together (5 min.)	Write three statements to summarize learning about unit rates (<i>SE p. 208</i>)
	5.7.3 Guided Practice (10 min.)	Practice using unit rates to solve real-world problems (<i>SE p. 206</i>)	5.7.6 Wrap-Up (~5 min.)	Formative assessment opportunity using unit rates for real-world problems (<i>SE p. 208</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A Additional Terms: N/A		N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may automatically think the larger quantity should be first when determining the unit rate. - Students may multiply when they should be dividing or divide when they should be multiplying. - Students may make computational errors if they have not mastered their multiplication and division of rational numbers.	- Consider allowing students to use a calculator. - Remind students that unit rates will always be rounded to the nearest hundredth when dealing with money. - Making a no-bake recipe in class would provide a kinesthetic entry point. - Comparing unit rates of products using online or newspaper flyers would reinforce the utility of finding the best value.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Think-Pair-Share Consider having students engage in 5.7.1 Warm-Up using the Think-Pair-Share routine. Initially, have students independently solve the problems. Then, have students discuss how they solved the problems and the answers they obtained with a partner. Have students resolve any differences in their answers through discussion. The discussion will help develop student understanding and confidence.	MA.K12.MTR.7.1: Real-World Contexts This lesson presents a multitude of opportunities for rich dialogue on real-world applications. The applications in this lesson include carnival ticket prices (5.7.1 Warm-Up), hiking and biking rates (5.7.2 Guided Instruction), rate of pay (5.7.3 Guided Practice), unit prices (5.7.4 Collaboration), and ratios of ingredients in a cheesecake recipe (5.7.6 Wrap-Up). Students use ratio tables and tape diagrams to display equivalent ratios and solve problems in these contexts.
	Consider placing students in groups and assigning groups to each number of the 5.7.4 Collaboration. By differentiating which groups engage with each portion of the component, the lesson can become a jigsaw in which students are responsible for sharing their learning with others. Jigsaw increases student agency by offering choice. Additionally, students are positioned as experts on their particular question, centering their approach and experience as expertise. Finally, the teacher could prompt students to share their partner's work rather than their own, further positioning students as analytical thinkers in processing and communicating the work of others.	
Lesson Notes:		

5.7.1 Warm-Up: Ticket Booth

Component Overview: Students are shown a poster of the price of different numbers of tickets at a festival and make predictions about which option is the best. Students then determine if there are any equivalent ratios on the poster and then find the factor by which the ratios need to be multiplied to find the second ratio.



5.7.1 Warm-Up: Ticket Booth

At a local fall festival, the following prices are shown at the ticket booth.

- Without performing any calculations, explain which option you think would be the best.

Answers will vary.

Ticket Prices	
1 Ticket	\$0.50
12 Tickets	\$5.00
25 Tickets	\$10.00
50 Tickets	\$25.00
120 Tickets	\$50.00

It depends on how many tickets you want. The unit rates are: 50 cents (if you buy 1 ticket), 0.41 $\bar{6}$ cents (if you buy 12 tickets), 40 cents (if you buy 25 tickets), 50 cents (if you buy 50 tickets), 0.41 $\bar{6}$ cents (if you buy 120 tickets). The best value is 25 tickets.

- Determine a pair of ratios, tickets:price, that appear to be equivalent. Write them in the space below.

If it is not possible to determine a pair of equivalent ratios from the sign, select one ratio and write another one that is equivalent.

Either of the following are correct:

1 ticket(s) : \$0.50

50 ticket(s) : \$25

These are both 50 cents per ticket.

12 ticket(s) : \$5

120 ticket(s) : \$50

These are both 0.41 $\bar{6}$ per ticket.



Ticket Booth – Token Economy

Consider utilizing a token economy for encouraging strong Mathematical Thinking and Reasoning (MTR) Standards throughout classroom discussion. When students analyze the thinking of others (MA.K12.MTR.4.1), make connections using patterns or structure (MA.K12.MTR.5.1), or utilize multiple representations when problem solving (MA.K12.MTR.2.1), the tickets can serve as positive reinforcement while also acting as an opportunity to apply learning of ratios when selecting prizes – like this Ticket Booth!

5.7.1 Warm-Up: Ticket Booth (continued)

3. By what factor can both values in the first ratio be multiplied to find the second ratio?

1 ticket(s) : \$0.50

50 ticket(s) : \$25

Multiply the number of tickets and cost both by 50.

12 ticket(s) : \$5

120 ticket(s) : \$50

Multiply the number of tickets and cost both by 10.

5.7.2 Guided Instruction: Interpreting Unit Rate

Component Overview: In this Guided Instruction, students review the definition of unit rate and apply this to identifying and finding the unit rates for several scenarios. Students then use tape diagrams to represent and interpret unit rates for different speeds, identifying which person is moving slower based on their unit rates.



5.7.2 Guided Instruction: Interpreting Unit Rate

A **unit rate** is a ratio comparing a number of units of a quantity to **1** unit of another quantity.

- Several rates are listed. Circle those that represent a unit rate.

earning \$42 for 3 hours

35 miles per gallon

75 miles per hour

2.5 miles in 24 minutes

each banana costs \$0.29

12 inches for every 1 foot

- Tia was hiking on the Florida National Scenic Trail. She hiked a total of 4.5 miles in 2 hours. The tape diagram models the distance and time she hiked.

Tia hiked 4.5 miles.

2.25 miles

2.25 miles

Tia hiked for 2 hours.

1 hour

1 hour

Notice that each tape diagram is the same size because they represent quantities in the same situation.

- Fill in each box of the tape diagram to show the value it represents.

See tape diagram.

- What is Tia's unit rate in miles per hour?

2.25 miles per hour



This component is intentionally designed to engage various learning styles. The tape diagrams provide a visual entry point for students and allow them to visualize and interpret different unit rates. Allow students to engage in verbal discourse to provide an auditory entry point.

5.7.2 Guided Instruction: Interpreting Unit Rate (continued)

- c. Draw a tape diagram that represents the time it would take Tia to hike 9 miles on the trail at this pace.

Tia hiked 9 miles.

2.25 mi.	2.25 mi.	2.25 mi.	2.25 mi.
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Tia hiked for 4 hours.

1 hr.	1 hr.	1 hr.	1 hr.
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This Guided Instruction is intentionally positioned to guide students through the representation of unit rates using tape diagrams. Allow students to work with a partner to create each tape diagram and circulate the room to observe how students are representing the unit rates. Review the representations as a class before releasing students to finish this component.

3. Carter bikes $12\frac{1}{2}$ miles in 5 hours around the city.

- a. Work with your partner to fill in each box of the tape diagram to show the value it represents. Be sure to include units.

2.5 miles	2.5 miles	2.5 miles	2.5 miles	2.5 miles
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1 hour	1 hour	1 hour	1 hour	1 hour
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- b. Interpret the unit rate for Carter's bike ride. Include units.

2.5 miles per hour

- c. Explain which person is moving slower, Tia or Carter. Include the unit rates in your explanation.

Tia is moving slower, because she travels a shorter distance (2.25 miles versus 2.5 miles) in the same amount of time (1 hour).

4. Complete the statement.

In each scenario, the unit rate in

☐ hours per mile
☐ miles per hour

 can

be determined by

☐ dividing
☐ multiplying

 the number of

☐ hours
☐ miles

traveled by the number of

☐ hours
☐ miles

 traveled.



Consider using talk moves like “revoice” or “reword” to give students an opportunity to synthesize their findings in their own words.

“In your own words, explain how you might use a tape diagram to determine a unit rate.”

“In your own words, how might you find the unit rate in a problem-solving situation?”

5.7.3 Guided Practice: Using Unit Rates to Solve Problems

Component Overview: Students revisit the ticket booth problem from 5.7.1 Warm-Up and use the unit rate of cost per ticket to complete a three-column table. Students then work with a partner to determine which ticket option is the best and which are equivalent and compare their answers to the Warm-Up. Partner pairs then use a table to answer more questions about the hourly rate of three employees.

5.7.3 Guided Practice: Using Unit Rates to Solve Problems

Unit rates would be helpful in considering the ticket booth from the Warm-Up. Miguel set up the table shown to consider the costs per ticket for each advertised price.

1. Work with your partner to complete the table. Then, answer the questions that follow.

Total Cost	Number of Tickets	Cost per Ticket
\$0.50	1	\$0.50
\$5.00	12	\$0.41 $\bar{6}$
\$10.00	25	\$0.40
\$25.00	50	\$0.50
\$50.00	120	\$0.41 $\bar{6}$

- a. Using the unit rates, explain which ticket option(s) is (are) the best deal.
The best deal is the 25 tickets for \$10.
- b. Using the unit rates, explain which ticket options are the same deal.
The 1 ticket for \$0.50 and the 50 tickets per \$25.00 are both \$0.50 per ticket.
The 12 ticket for \$5.00 and the 120 tickets per \$50.00 are both \$0.41 $\bar{6}$ per ticket.
- c. Compare your answer to the previous question to the equivalent ratios you determined in the Warm-Up. Make any revisions to your previous thinking, as necessary.
- d. With your partner, discuss recommendations for how the high school should change its prices for carnival tickets.



Notice the partner work in this component. Be intentional with grouping, possibly using data or informal observations from the Lesson 6 Wrap-Up to inform pairing.



Scaffold this component with intentional questions like:

- Was it a better value to buy more or fewer tickets? How do you know?
- Based on the ticket prices and unit rate, how can the high school change its prices for carnival tickets? Why would this be helpful?



Prompt students toward discussing unit price and its role in determining lower prices in bulk. Normally larger amounts are priced cheaper per unit – just like the cost of most items at the store is cheaper per unit the more you buy, so is the cost per ticket.

5.7.3 Guided Practice: Using Unit Rates to Solve Problems (continued)

2. Holly, Frank, and Ella work part-time for Shoes-R-Us. The table shows how much money each person earns during a work week.

Employee	Weekly Pay	Number of Hours Worked
Holly	\$165	20
Frank	\$162	18
Ella	\$170	20

Work with your partner to answer the following questions using the information in the table.

- Find and interpret the hourly rate Holly earns.
Holly earns \$8.25 per hour. This means that each hour she earns \$8.25.
- Which employee earns the higher hourly rate, Ella or Frank?
Frank earns \$9.00 per hour. Ella earns \$8.50 per hour. Therefore, Frank earns the higher hourly rate.
- If Frank works 25 hours next week, how much should he expect to earn?
Frank will earn \$225.00 after working 25 hours.



Encourage student discourse throughout this component as they work in partner pairs. Highlight and surface whole group any student responses that connect hourly rate as equivalent to unit rate because it is per 1 hour.

5.7.4 Collaboration: Using and Interpreting Unit Rates

Component Overview: Students collaboratively interpret and use unit rates to solve real-world problems.



5.7.4 Collaboration: Using and Interpreting Unit Rates

Work with your partner to interpret the unit rates for each situation.

1. Tyler paid \$12 for 4 large candy bars.
 - a. Tyler interprets the unit rate as $\frac{\$3}{1 \text{ bar}}$.
His friend Ben interprets the unit rate as $\frac{1 \text{ bar}}{\$3}$. Which person interpreted the unit rate correctly?
Both Tyler and Ben are correct.
 - b. Based on the unit rate, how many candy bars can Tyler purchase for \$45?
Tyler can purchase 15 candy bars for \$45.00.
2. Aaliyah's family drinks a total of 10 gallons of milk every 6 weeks and 4 gallons of apple juice every 2 weeks.
 - a. Which beverage does Aaliyah's family drink more of in one week?
Aaliyah's family drinks more apple juice (2 gallons) than milk ($1\frac{2}{3}$ gallons) in one week.
 - b. How many weeks does it take the family to consume 1 gallon of milk?
It takes the family $\frac{6}{10} = \frac{3}{5}$ of a week (or 4.2 days) to consume 1 gallon of milk.
3. Carlos buys 24 bottles of spring water for \$3.60 and 3 jars of pasta sauce for \$9.69.
 - a. What does each item, a bottle of spring water and a jar of pasta sauce, cost per unit?
The spring water costs \$0.15 per bottle. The pasta sauce costs \$3.23 per jar.
 - b. What would be Carlos' total cost if he purchased 10 bottles of spring water and 2 jars of pasta sauce at their unit prices? *Ten bottles of spring water would cost \$1.50. Two jars of pasta sauce would cost \$6.46. The total cost would be \$7.96.*



Consider positioning students to discuss question prompts rather than writing their answers. This entry point might engage auditory learners. Use sentence stems to encourage conversation such as "To find the unit rate, I..."



Consider using a jigsaw approach for engaging students.

- Question 1: Lower entry point with one-step multiplication and division
- Question 2: Accessible on grade level involving operations with fractions
- Question 3: Higher entry point for students who should engage with decimal operations

Students should then find a partner with a different number and explain both their process and product. The teacher should then call on students to explain the work of their partner, further extending the mathematical thinking of self and others.

5.7.5 Bring It Together: Interpreting Unit Rates

Component Overview: Students work with a partner to consolidate their learning by writing three statements to summarize learning about unit rates. Consider reviewing 2 or 3 statements with the class before moving on if time permits.



5.7.5 Bring It Together: Interpreting Unit Rates

Work with your partner to write three statements to summarize your learning about unit rates from today's lesson. *Answers may vary.*

1. *I learned that to find a unit rate, you divide both quantities by the value of one of the quantities.*
2. *I learned that for unit rates expressing cost per unit, the smallest unit rate is considered the best value.*
3. *I learned that the unit rate can be used to determine other equivalent ratios.*



Consider sentence stems if you are targeting specific learning goals or consider having the students answer each of the three Essential Questions as their statements.

5.7.6 Wrap-Up: Ticket Out the Door

Component Overview: Students demonstrate their understanding of interpreting and using unit rates to solve real-world problems.



5.7.6 Wrap-Up: Ticket Out the Door

A cheesecake recipe says, "Mix 12 ounces (oz) of cream cheese with 15 oz of sugar."

1. How many ounces of cream cheese are needed per ounce of sugar?

According to the recipe, $\frac{12}{15} = \frac{4}{5} = 0.8$ ounces of cream cheese are needed per ounce of sugar.

2. How many ounces of sugar are used for each ounce of cream cheese?

According to the recipe, $\frac{15}{12} = \frac{5}{4} = 1.25$ ounces of sugar are needed per ounce of cream cheese.

3. Max plans to make more than one cheesecake. How much cream cheese will she need to mix with 35 oz of sugar?

Max will need 28 ounces of cream cheese to mix with 35 ounces of sugar.



Prompt students to create an additional representation of any of the questions in 5.7.6 Wrap-Up. Their choice of double number line or table will provide additional points of data when creating small groups in the next lesson.